SIMATS ENGINEERING

ASSESSMENT TOOL 1:

Experiments

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA1520

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.* (BL3)

Dave's Taxonomy: Precision (Level 3) **Total Marks: 40**
Question Count: 4

Code of Conduct

I _____G. Yashmita (192511032)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

Set 3- Assessment Tasks

Experiment

Experiment 1: Create a simple cloud software application for Mark Sheet Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as name of the student, Reg no of the student , Subjects , Marks, Average, percentage, rank, overall performance etc.

Experiment 2: Create a simple cloud software application for Property Buying & Rental process (In Chennai city) using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Buy, Rent, commercial, Rental Agreement , Rental Agreement, Area, place, Range, property value etc.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 5 CPU, 12 GB RAM and 8 GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Snapshot of a VM and Test it by loading the Previous Version/Cloned VM.

Experiment 1: Mark Sheet Processing System

Aim:

To develop a cloud-based Mark Sheet Processing System using Zoho Creator to store student marks and calculate total marks, average, percentage, and performance.

Software Required:

• Zoho Creator

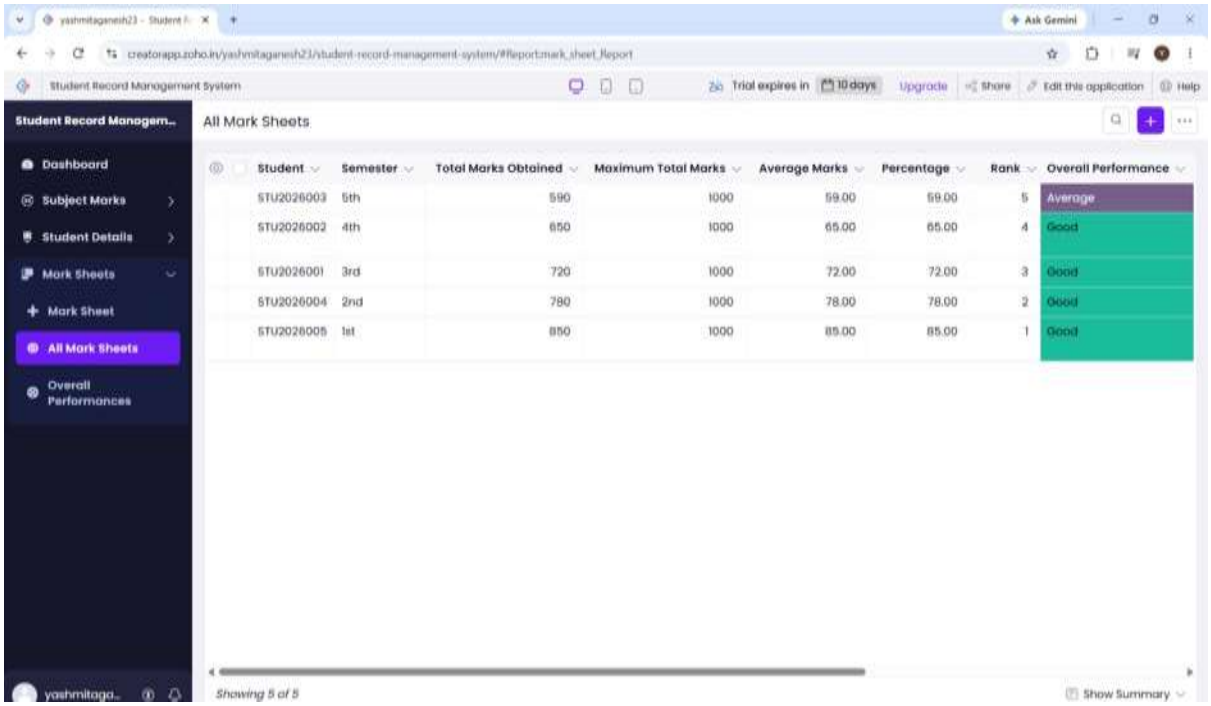
• Web browser

• Internet connection

Procedure:

1. Create a new Zoho Creator application named Mark Sheet Processing System using the AI application generator.
2. Create a Student Details form containing Student ID, Name, Department, Year, and Register Number.
3. Create a Subject Marks form to store subject names and marks obtained by students.
4. Create a Mark Sheet form to display student details, subject marks, total marks, average, percentage, and result.
5. Configure calculations for total marks, average, and percentage wherever supported.
6. Generate reports to display student marks and academic performance.
7. Create a dashboard to summarize student results and enter sample student records to test the application.

Execution:



Result:

Thus, the Mark Sheet Processing System was created using Zoho Creator to maintain student marks and process academic performance details.

Experiment 2: Chennai Property Buying and Rental System using SaaS

Aim: To develop a cloud-based Property Buying and Rental System for Chennai using Zoho Creator, supporting property listings and customer enquiries.

Software Required:

• Zoho Creator

• Web browser

• Internet connection

Procedure:

1. Log in to Zoho Creator and create an application named Chennai Property Management System.

2. Create a Property Listing form with fields such as property ID, title, property type, transaction type, locality, address, price or rent, bedrooms, area, and owner name.

3. Configure the transaction type to support buying and renting.

4. Create a Customer Enquiry form containing customer details, phone number, email, preferred locality, budget, preference, enquiry date, and message.

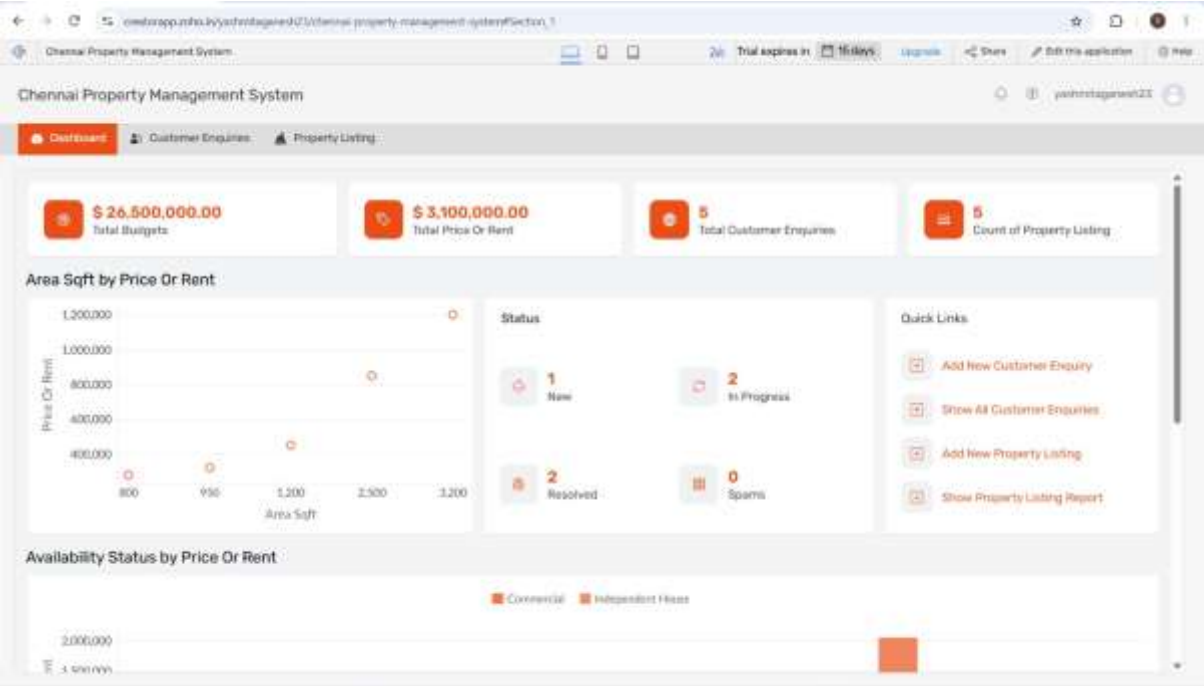
5. Create reports to display property listings and customer enquiries.

6. Design a dashboard containing summary cards, charts, and quick links.

7. Add sample property and enquiry records.

8. Open the application and verify the dashboard and reports.

Execution:



Result: The Chennai Property Buying and Rental System was created using Zoho Creator with property listings, customer enquiries, reports, and a dashboard.

Experiment 3: Demonstrate Virtualization Using a Type-2 Hypervisor

Aim:

To demonstrate virtualization by installing a Type-2 hypervisor and creating and configuring a virtual machine with a guest operating system.

Software Required:

• VMware Workstation

• Ubuntu Linux ISO image

• Windows host operating system

• Sufficient RAM and disk space

Procedure:

1. Download and install VMware Workstation on the host computer.

2. Open VMware Workstation and select the option to create a new virtual machine.

3. Select the installation media containing the Ubuntu Linux ISO image.

4. Configure the virtual machine's name and storage location.

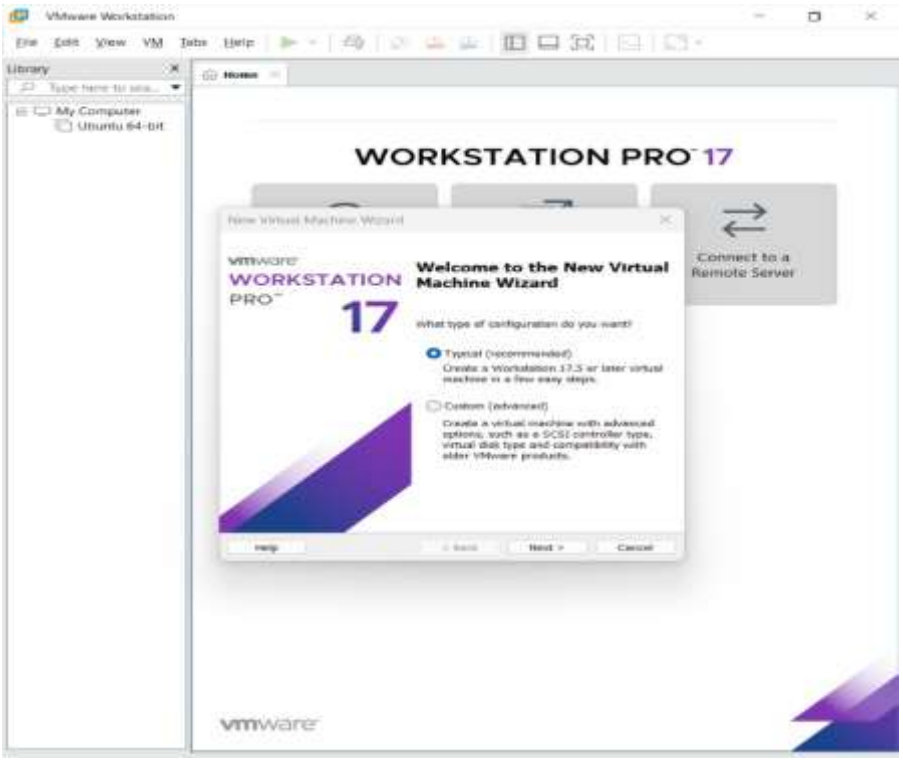
5. Allocate the required processor, memory, and disk resources.

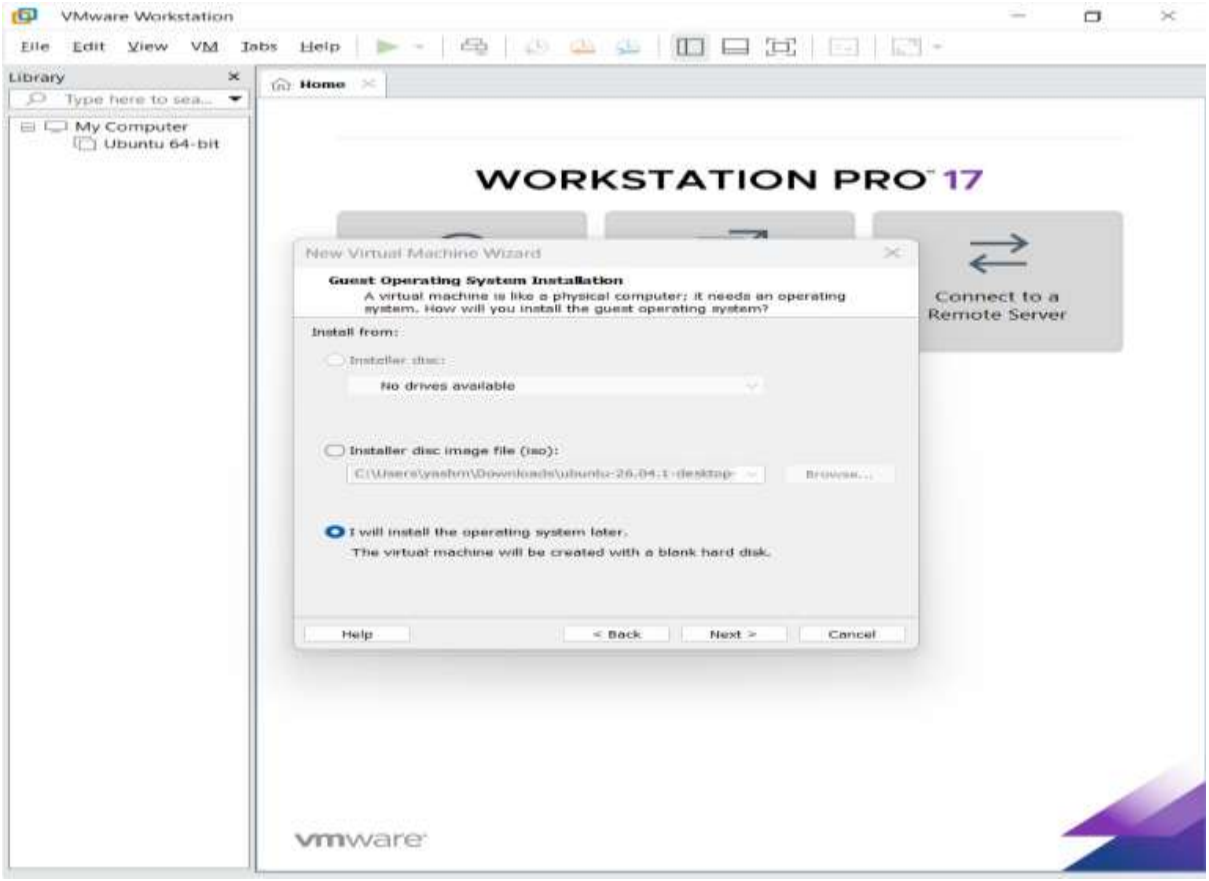
6. Start the virtual machine and install Ubuntu Linux.

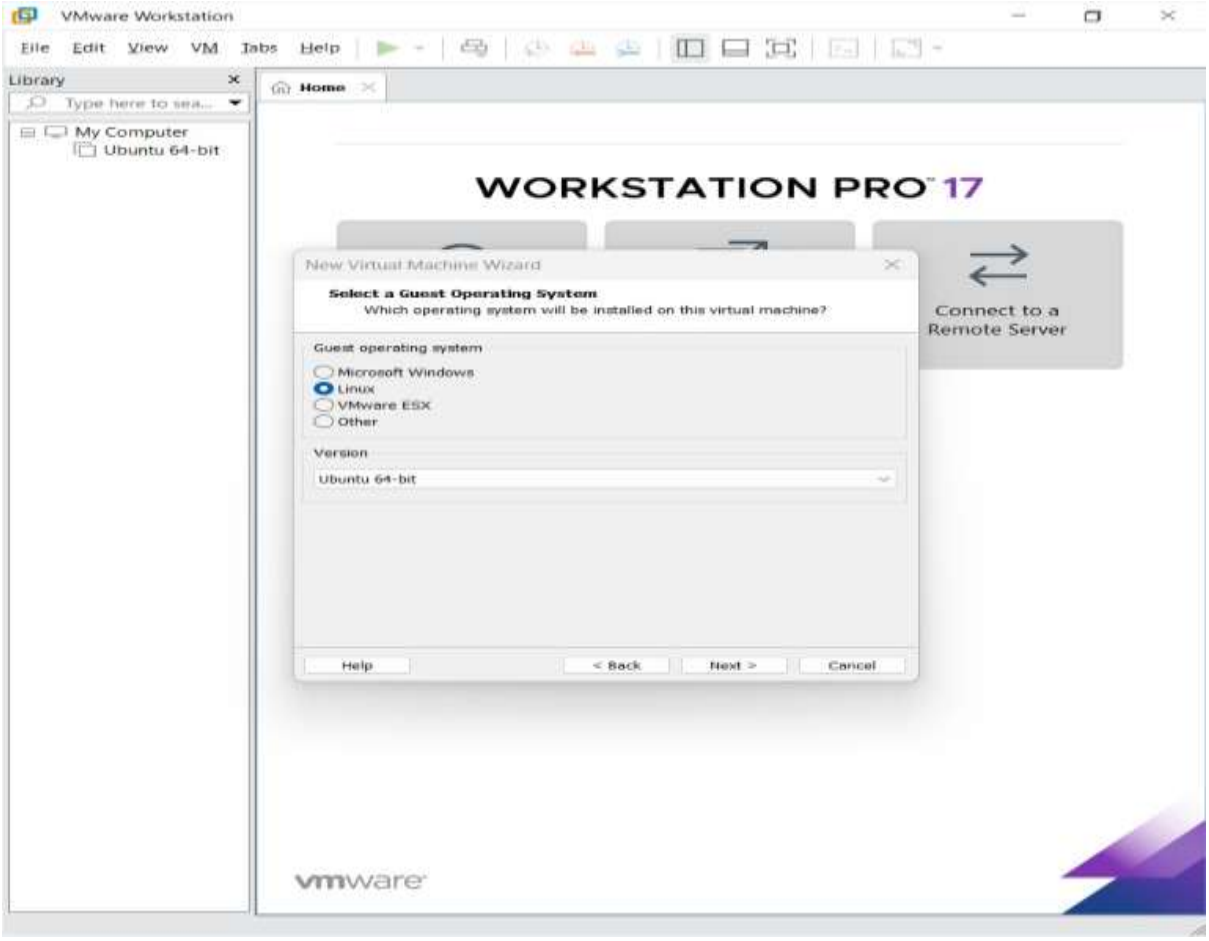
7. Complete the operating system setup.

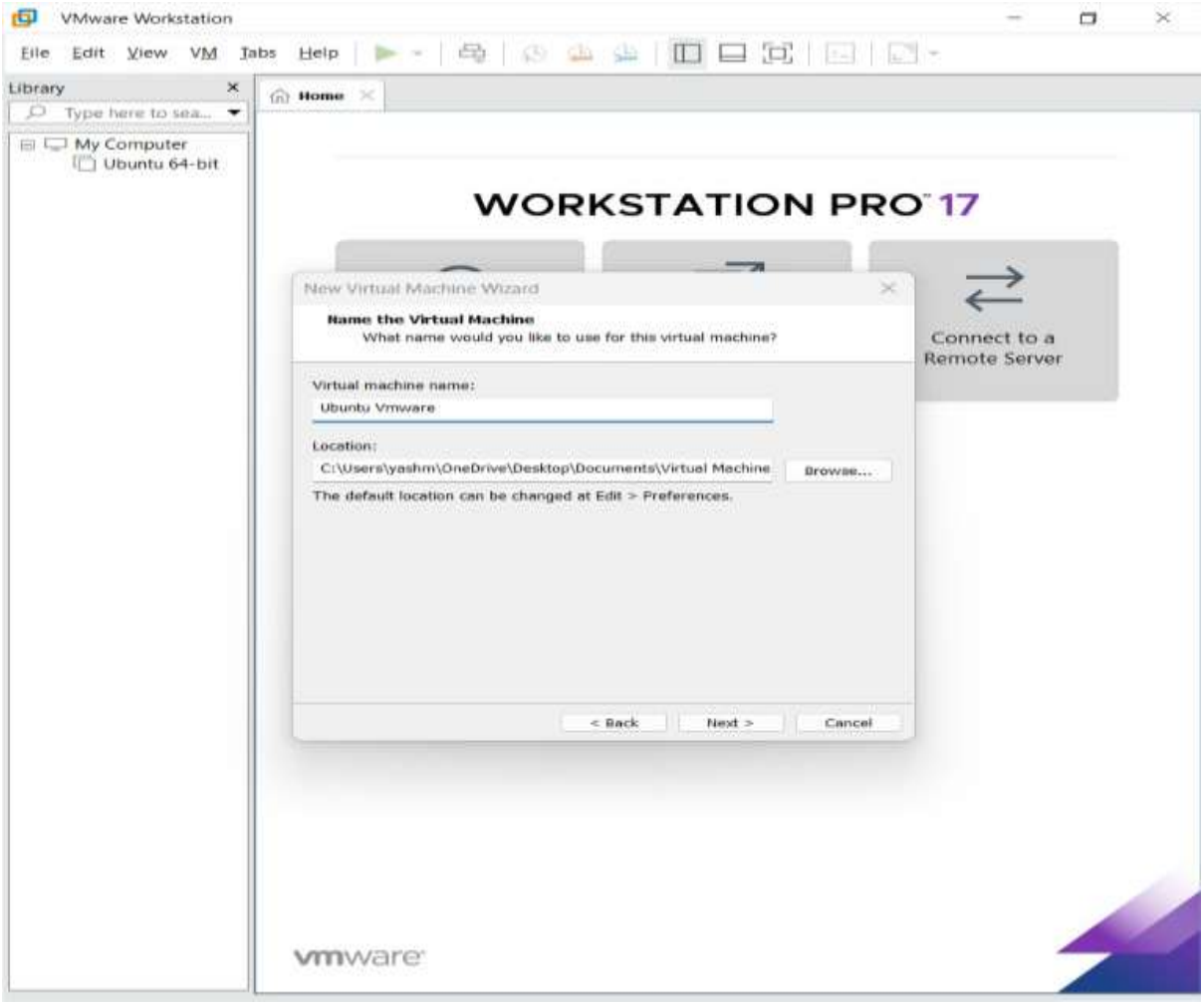
8. Start the virtual machine and verify that Ubuntu runs successfully.

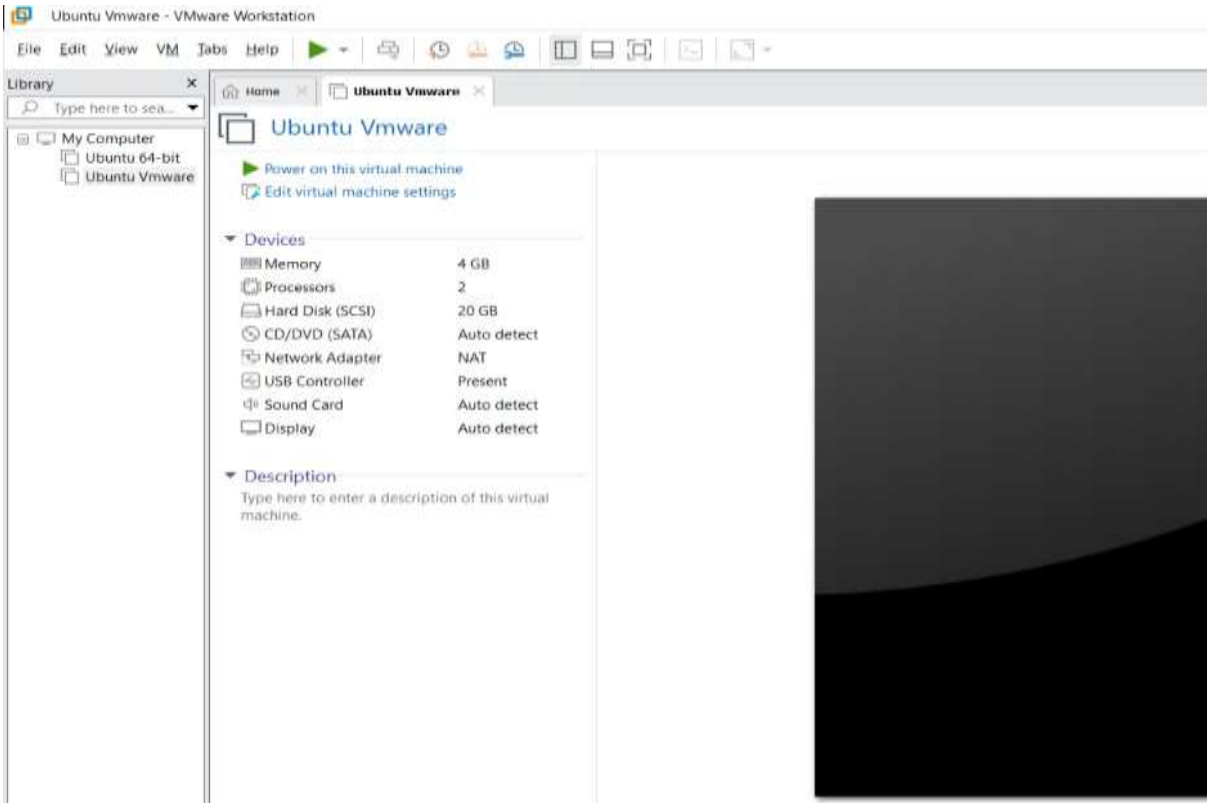
Execution:

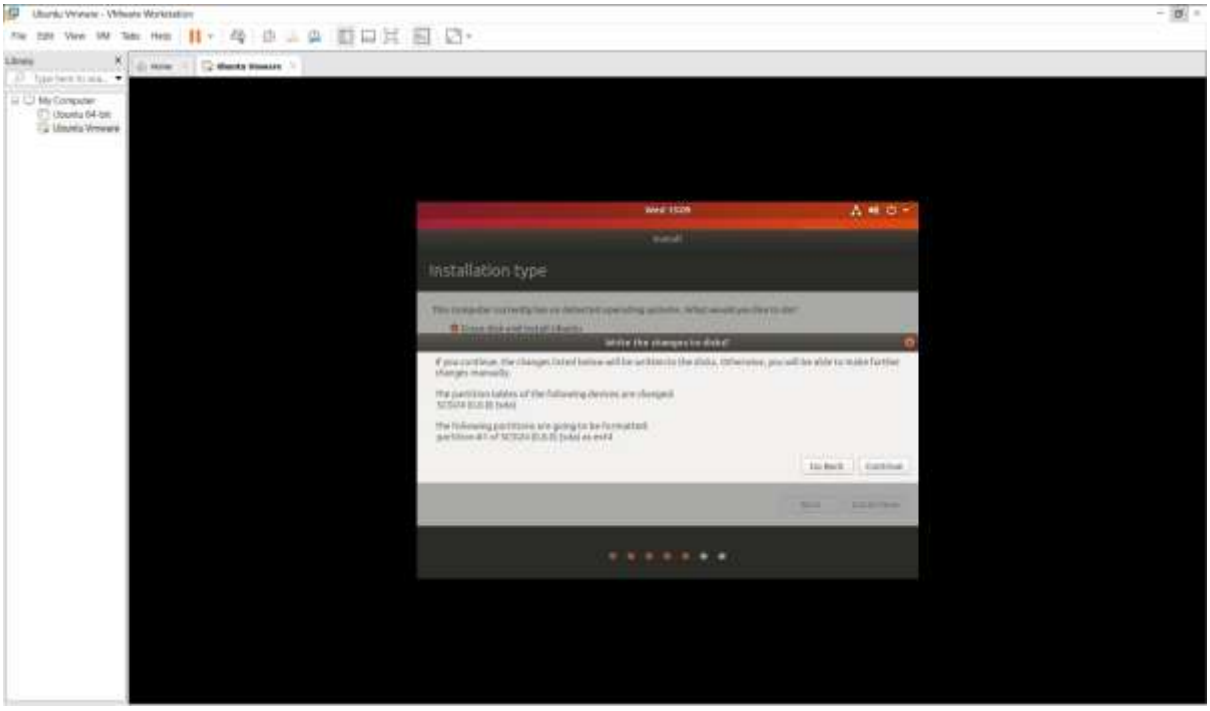


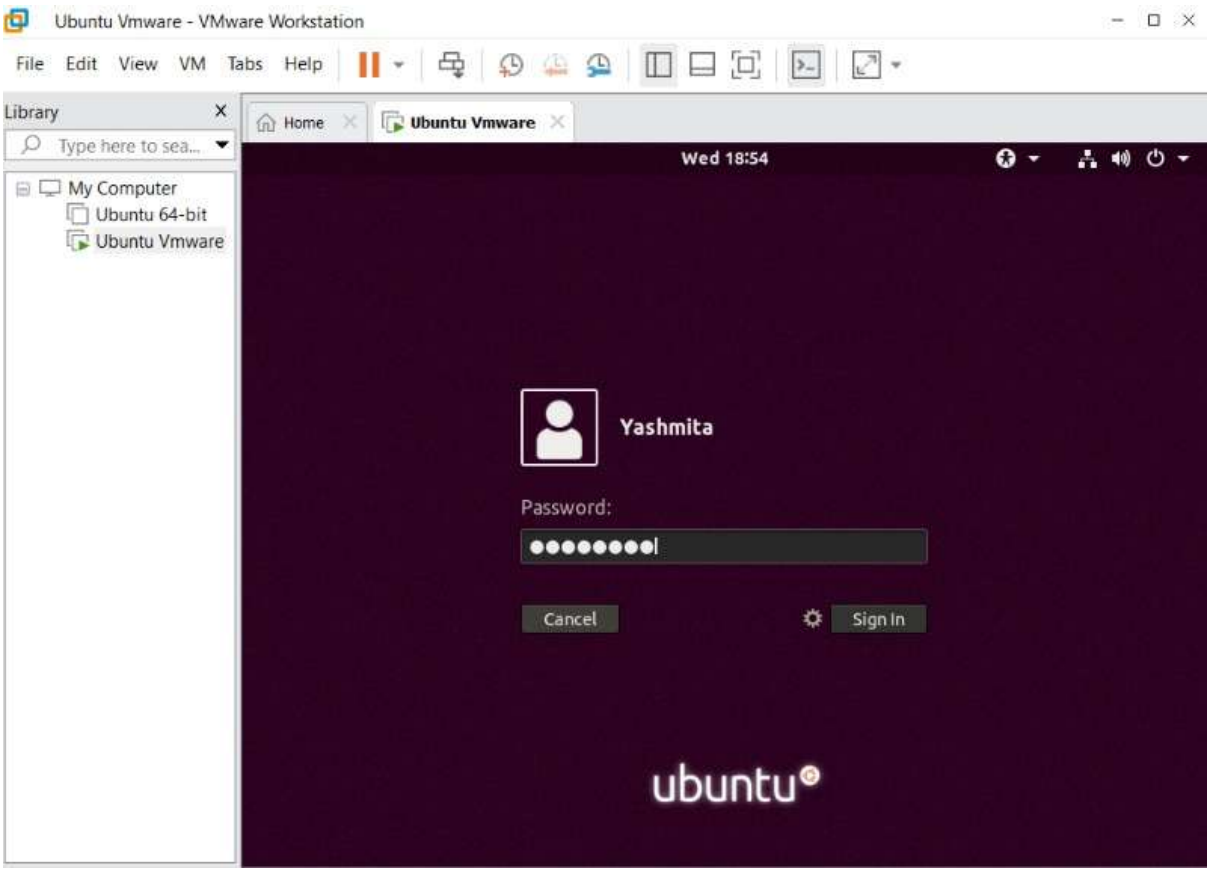

VMware Workstation
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
Ubuntu 64-bit
Home
WORKSTATION PRO 17
New Virtual Machine Wizard
Guest Operating System Installation
A virtual machine is like a physical computer; it needs an operating system. How will you install the guest operating system?
Install from:
Installer disc:
No drives available
Installer disc image file (iso):
C:\Users\yashm\Downloads\ubuntu-26.04.1-desktop
Browse...
I will install the operating system later.
The virtual machine will be created with a blank hard disk.
Help
< Back
Next >
Cancel
Connect to a Remote Server
vmware

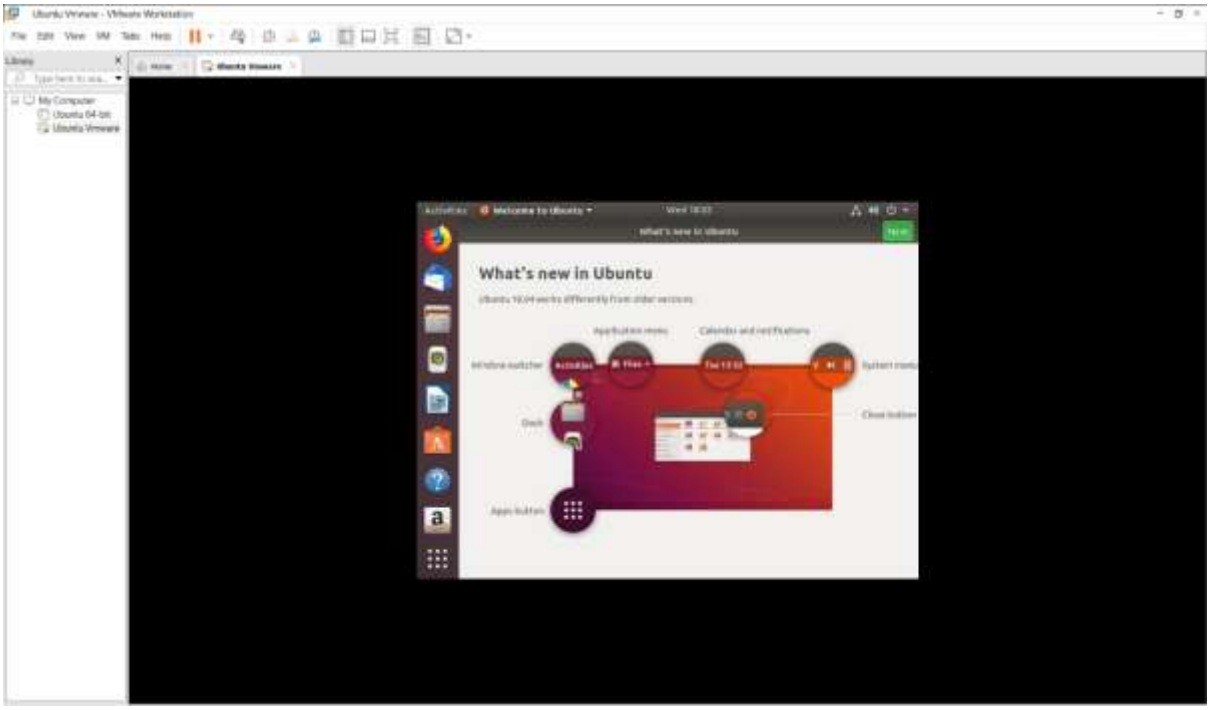

VMware Workstation
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
Ubuntu 64-bit
Home
WORKSTATION PRO 17
New Virtual Machine Wizard
Select a Guest Operating System
Which operating system will be installed on this virtual machine?
Guest operating system
Microsoft Windows
Linux
VMware ESX
Other
Version
Ubuntu 64-bit
Help
< Back
Next >
Cancel
Connect to a Remote Server
vmware


VMware Workstation
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
Ubuntu 64-bit
Home
WORKSTATION PRO 17
Connect to a Remote Server
New Virtual Machine Wizard
Name the Virtual Machine
What name would you like to use for this virtual machine?
Virtual machine name:
Ubuntu Vmware
Location:
C:\Users\yashm\OneDrive\Desktop\Documents\Virtual Machine
Browse...
The default location can be changed at Edit > Preferences.
< Back
Next >
Cancel
vmware


Ubuntu Vmware - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
Ubuntu 64-bit
Ubuntu Vmware
Home
Ubuntu Vmware
Ubuntu Vmware
Power on this virtual machine
Edit virtual machine settings
Devices
Memory 4 GB
Processors 2
Hard Disk (SCSI) 20 GB
CD/DVD (SATA) Auto detect
Network Adapter NAT
USB Controller Present
Sound Card Auto detect
Display Auto detect
Description
Type here to enter a description of this virtual machine.


Installation type


Ubuntu Vmware - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
Ubuntu 64-bit
Ubuntu Vmware
Home
Ubuntu Vmware
Wed 18:54
Yashmita
Password:
Cancel
Sign In
ubuntu



Result: Virtualization was demonstrated by installing and configuring a guest operating system using VMware Workstation, a Type-2 hypervisor.

Experiment 4: Create and Test a Virtual Machine Snapshot

Aim:

To create a snapshot of a virtual machine and verify that the virtual machine can be restored to its previous state.

Software Required:

• VMware Workstation

• An existing configured virtual machine

• Guest operating system

Procedure:

1. Open VMware Workstation and select the configured virtual machine.

2. Start the virtual machine and verify that the operating system works correctly.

3. Open the Snapshot menu and select the option to take a snapshot.

4. Enter a suitable snapshot name and description.

5. Confirm the snapshot creation.

6. Make a small, reversible change to the guest operating system, such as creating a test file.

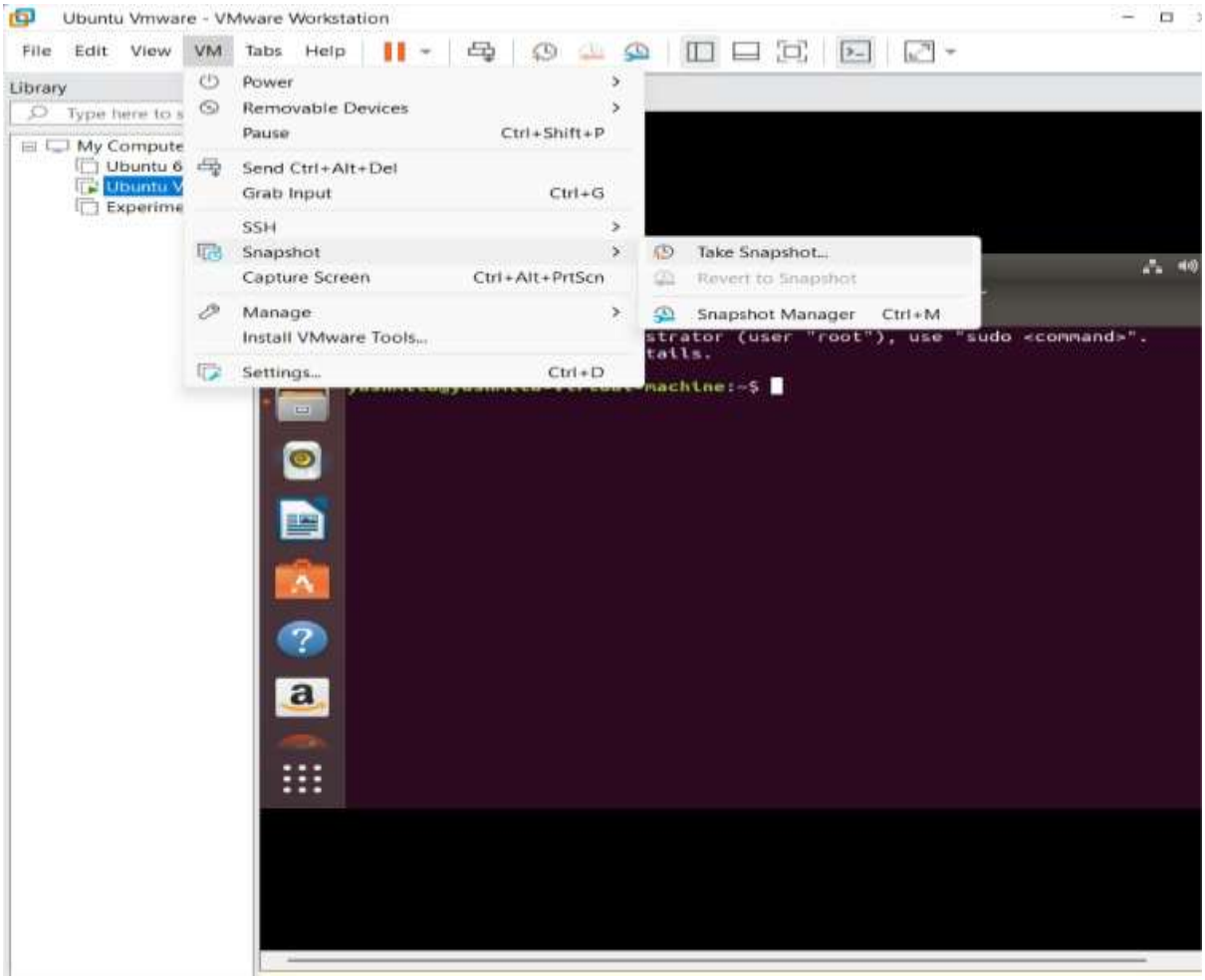
7. Open the Snapshot Manager.

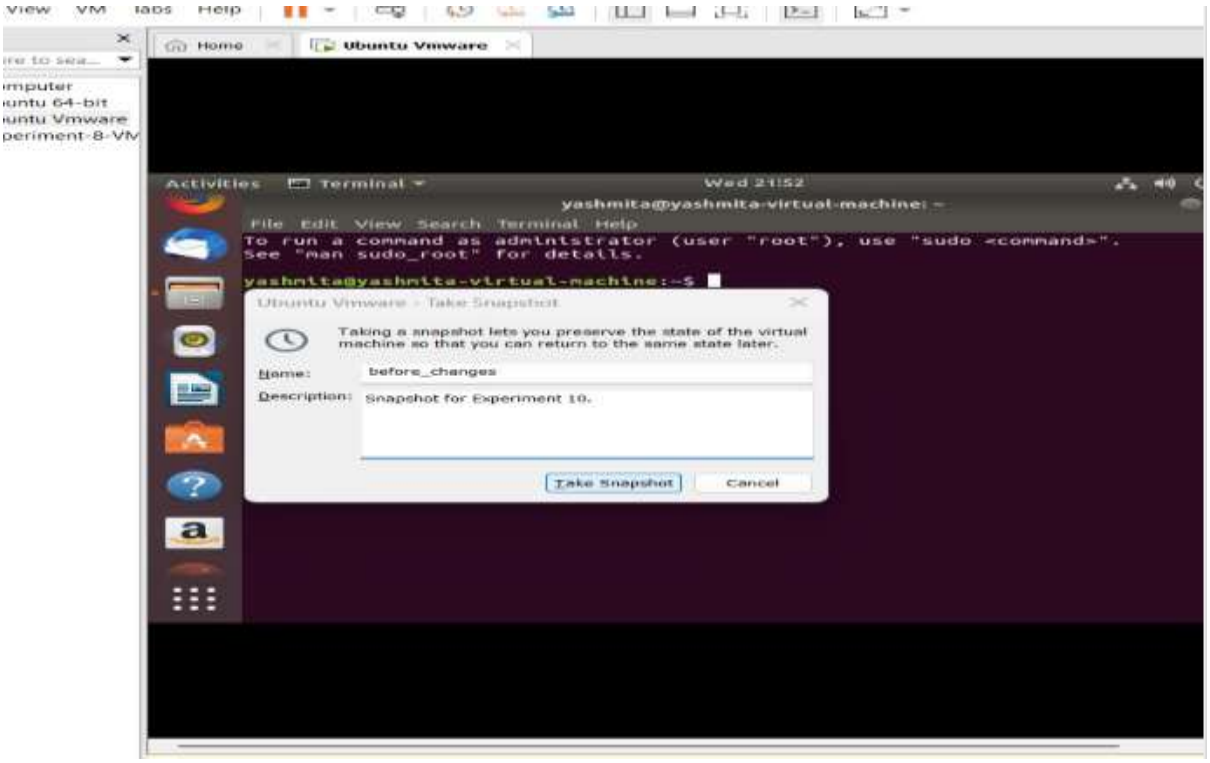
8. Select the previously created snapshot and choose the option to revert to it.

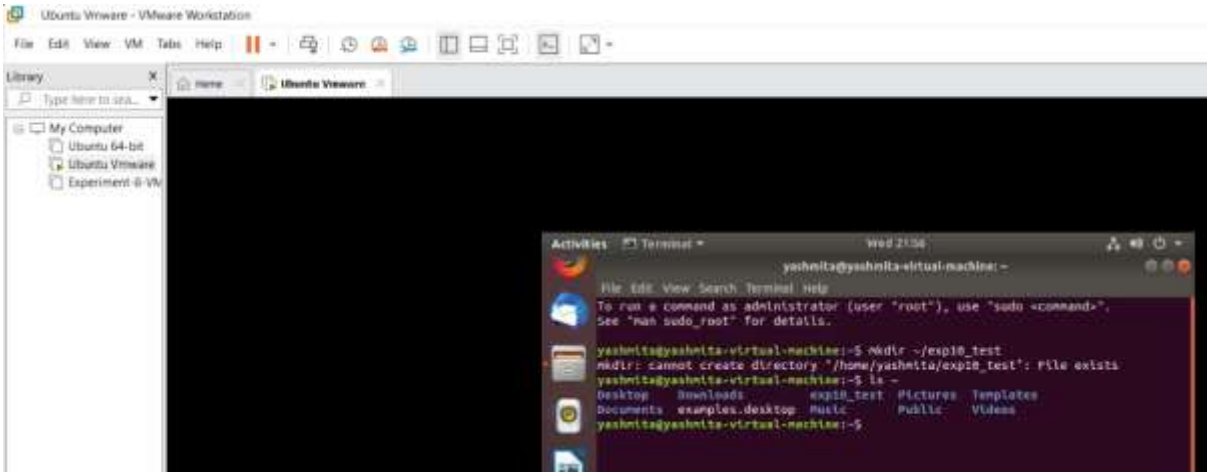
9. Confirm the restoration and start or resume the virtual machine.

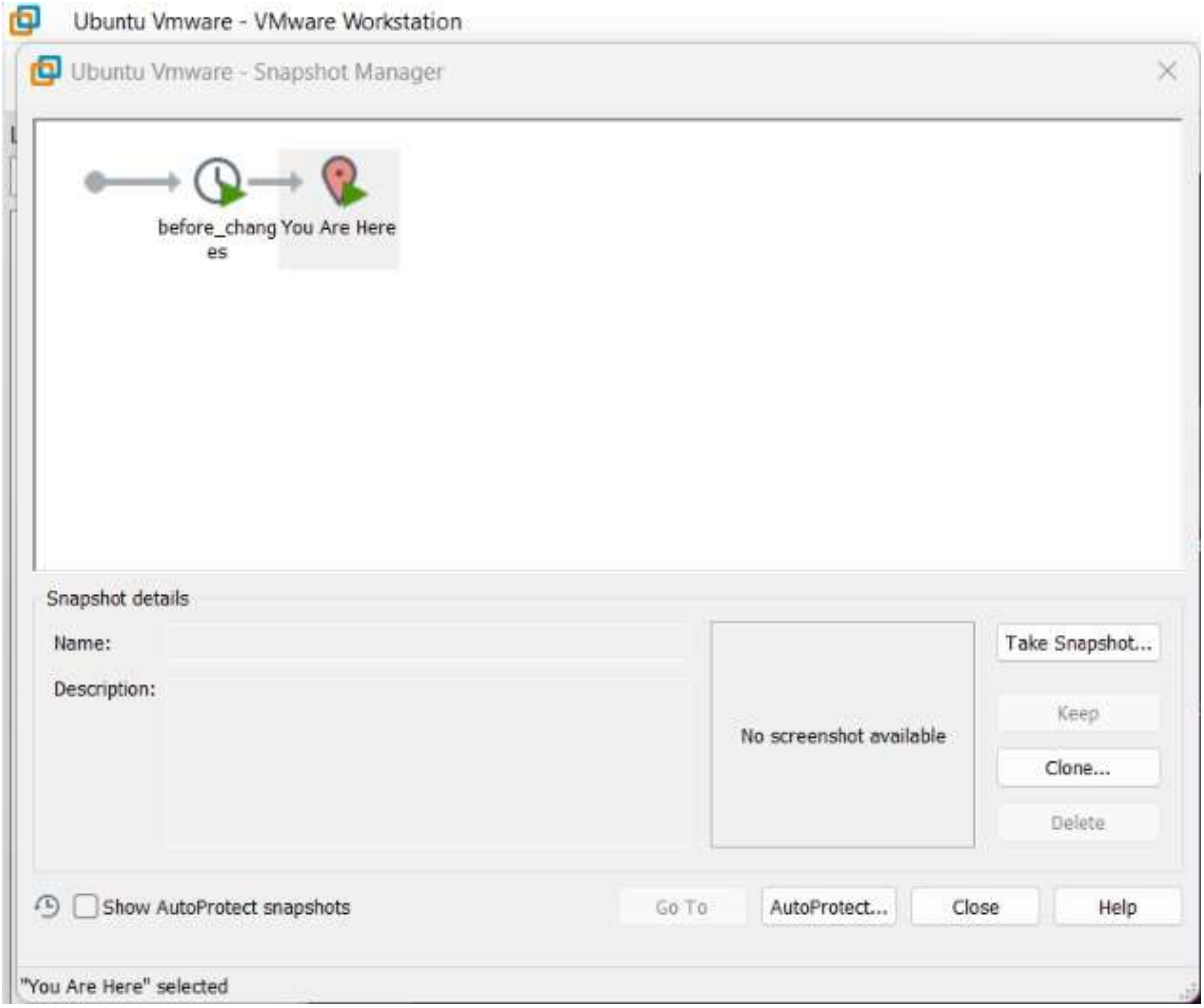
10. Verify that the virtual machine has returned to the saved state.

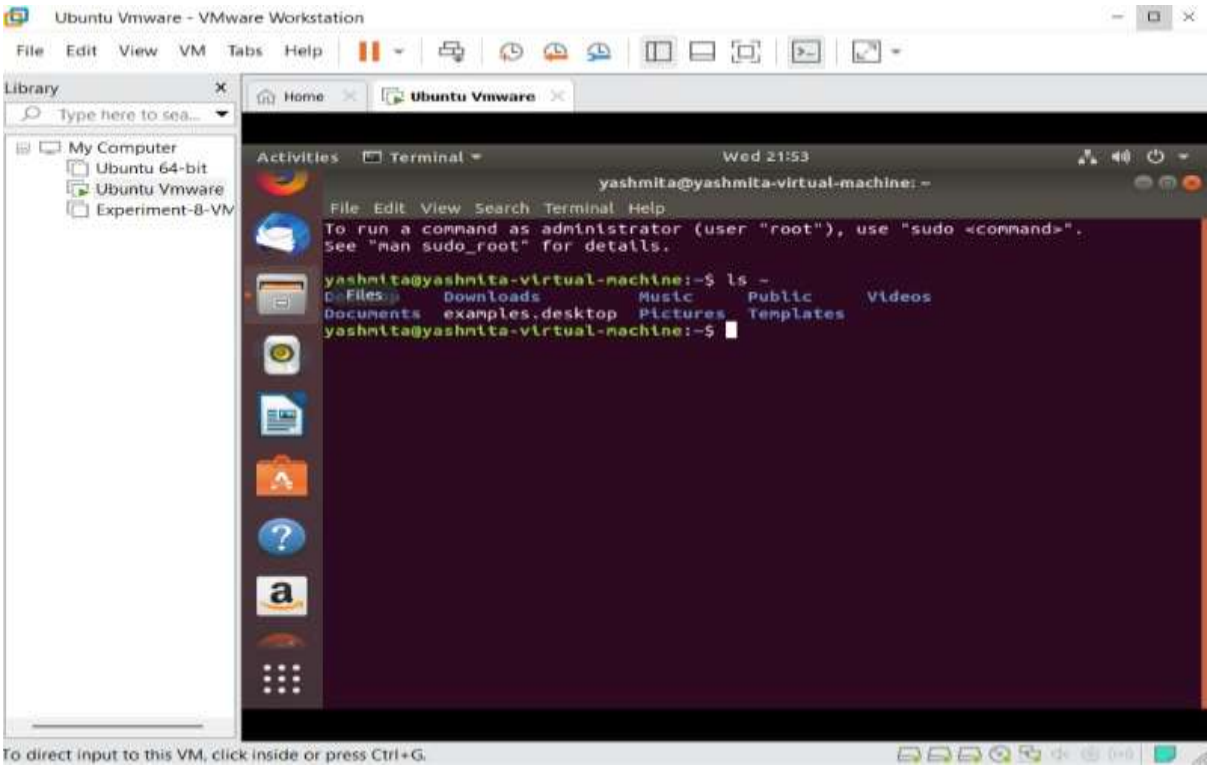
Execution:




View VM Tabs Help
Home
Ubuntu Vmware
Activities
Terminal
Wed 21:52
yashmita@yashmita-virtual-machine: ~
File Edit View Search Terminal Help
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
yashmita@yashmita-virtual-machine:~$
Ubuntu Vmware - Take Snapshot
Taking a snapshot lets you preserve the state of the virtual machine so that you can return to the same state later.
Name:
before_changes
Description:
Snapshot for Experiment 10.
Take Snapshot
Cancel


Ubuntu Vmware - VMware Workstation
File Edit View VM Tabs Help
Library
My Computer
Ubuntu 64-bit
Ubuntu Vmware
Experiment-8-VM
Home
Ubuntu Vmware
Activities
Terminal
Wed 21:56
yashmita@yashmita-virtual-machine: ~
File Edit View Search Terminal Help
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
yashmita@yashmita-virtual-machine:~$ mkdir ~/exp10_test
mkdir: cannot create directory '/home/yashmita/exp10_test': File exists
yashmita@yashmita-virtual-machine:~$ ls ~
Desktop Downloads exp10_test Pictures Templates
Documents examples.desktop Music Public Videos
yashmita@yashmita-virtual-machine:~$


Ubuntu Vmware - VMware Workstation
Ubuntu Vmware - Snapshot Manager
before_changes
You Are Here
Snapshot details
Name:
Description:
No screenshot available
Take Snapshot...
Keep
Clone...
Delete
Show AutoProtect snapshots
Go To
AutoProtect...
Close
Help
"You Are Here" selected


Ubuntu Vmware - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
Ubuntu 64-bit
Ubuntu Vmware
Experiment-8-VM
Home
Ubuntu Vmware
Activities
Terminal
Wed 21:53
yashmita@yashmita-virtual-machine: ~
File Edit View Search Terminal Help
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
yashmita@yashmita-virtual-machine:~$ ls
Desktop Downloads Music Public Videos
Documents examples.desktop Pictures Templates
yashmita@yashmita-virtual-machine:~$
To direct input to this VM, click inside or press Ctrl+G.

Result: A virtual machine snapshot was created and used to restore the machine to a previous state.